

6. Legos® The time required for kindergarten children to assemble a specific Lego creation was measured for children who had been instructed for four different lengths of time. Five children were randomly assigned to each instructional group. The length of time (in minutes) to assemble the Lego creation was recorded for each child in the experiment.

Training Period (hours)				
.5	1.0	1.5	2.0	
8	9	4	4	
14	7	6	7	
9	5	7	5	
12	8	8	5	
10	9	6	4	

Use the Kruskal-Wallis H -Test to determine whether there is a difference in the distribution of times for the four different lengths of instructional time. Use $\alpha = .01$.

Rank

	.5	1.0	1.5	2.0
1	13	16	2	2
2	20	10	7.5	10
3	16	5	10	5
4	19	13	13	5
5	18	16	7.5	2

H_0 : the distribution of four trainings are same

H_a : at least one distribution is different

$$\alpha = 0.01 \quad n = 5 \times 4 = 20$$

$$H_{STAT} = \frac{12}{n(n+1)} \cdot \sum \frac{(\sum U_i)^2}{n_i} - 3(n+1)$$

$$H^* = \frac{12}{20(20+1)} \cdot \left(\frac{86^2}{5} + \frac{60^2}{5} + \frac{40^2}{5} + \frac{24^2}{5} \right) - 3 \cdot 21$$

$$= 12.2686$$

$$\chi^2_{3, 0.01} = 11.3449$$

$$RR: \{H \mid H > 11.3449\} \quad 12.2686 > 11.3449$$

$\Rightarrow$ Reject H_0

There is sufficient evidence to say that at least one distribution is different.